For this one I am going with Kubernetes as my cloud design tool, and the case for it really comes down to portability. Most of the vendor specific options, things like Azure Resource Manager or AWS CloudFormation, are excellent for provisioning resources but they lock the design into that one provider's syntax and ecosystem. Kubernetes takes a different approach entirely, it defines applications and infrastructure through a declarative, open standard that runs the same way whether it sits on AWS, Azure, GCP, or even on premises hardware (Nigel and Pushkar, 2017).

That portability matters a lot more once you are actually working across providers rather than just picking one and staying there. Natarajan and Jeveen (2024) make this point clearly in their multi cloud handbook, showing that organisations increasingly design applications to run across more than one cloud, whether for resilience, pricing leverage, or avoiding being stuck with a single vendor, and Kubernetes is basically the common language that makes that realistic rather than a huge maintenance burden.

Where I think Kubernetes genuinely pulls ahead of the alternatives is in how mature its ecosystem has become around actually operating what you have designed, not just deploying it once. Autoscaling, self healing, rolling updates and a massive base of existing tooling all come as part of the platform rather than being bolted on afterwards. That said, it is worth being honest that this flexibility comes with a cost awareness problem, since declarative infrastructure defined this broadly can make it genuinely difficult to see where cost is actually being generated across a sprawling set of manifests (Feitosa et al., 2023). Even with that caveat, I still think the trade off favours Kubernetes for anyone designing cloud native systems that need to outlive a single provider relationship.

References

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